

EXPLAINABLE ARTIFICIAL INTELLIGENCE

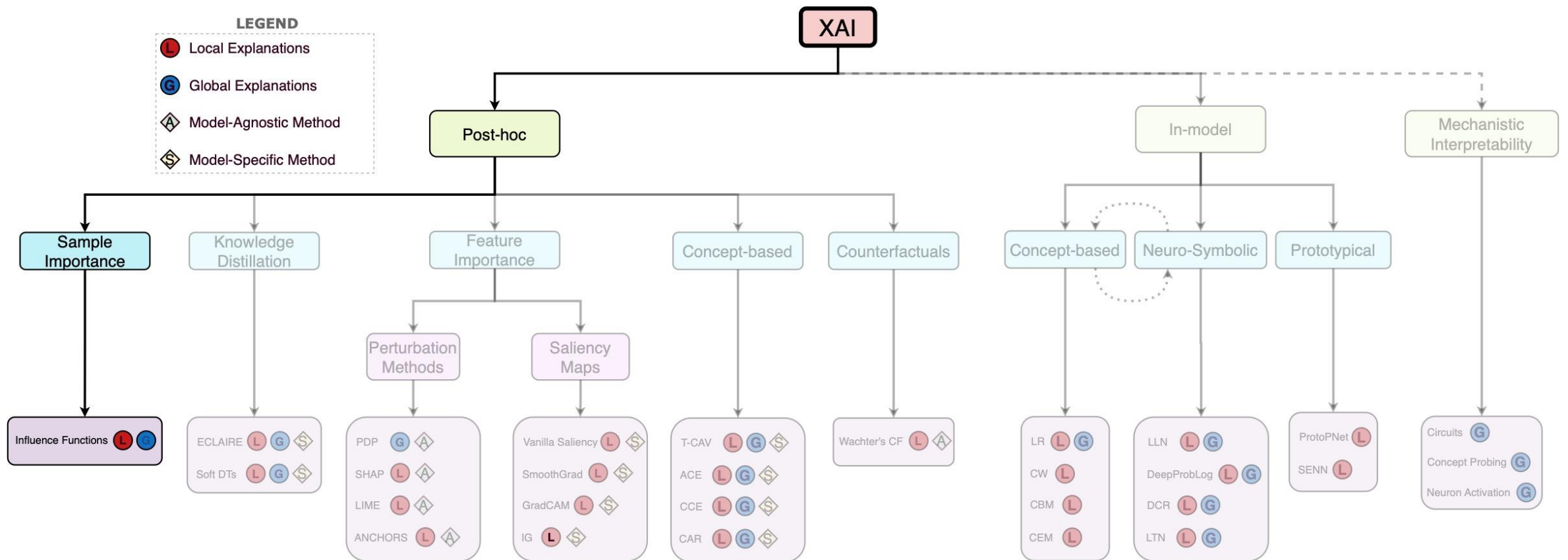
L193 – Lecture 5 – Lent 2025



UNIVERSITY OF
CAMBRIDGE



WHERE TO GO NEXT?



INFLUENCE FUNCTIONS



INFLUENCE FUNCTIONS: MOTIVATION

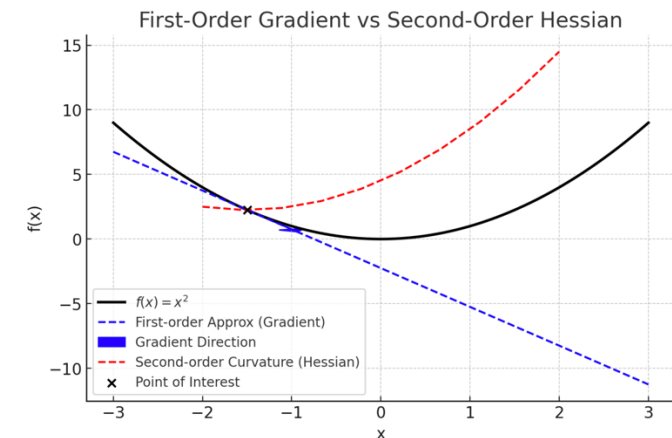
- Feature/concept importance vs training point importance
- How can we do this?
 - **Re-train** the model with each training point removed 
 - **Approximating the effects of removal** of a training point using **influence functions (IF)**

INFLUENCE FUNCTION: FORMULATION I

Result dating back to 1982 [1]: **Influence of up-weighting x on the parameters θ can be calculated using the inverse Hessian:**

$$I_{\text{up,params}}(x) \stackrel{\text{def}}{=} \left. \frac{d\hat{\theta}_{\epsilon,x}}{d\epsilon} \right|_{\epsilon=0} = -H_{\hat{\theta}}^{-1} \nabla_{\theta} L(x, \hat{\theta}) \quad H_x = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_1 \partial x_2} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix}$$

Second derivative measures (curvature)



[1] Cook, R. D. and Weisberg, S. Residuals and influence in regression. New York: Chapman and Hall, 1982.

INFLUENCE FUNCTION: FORMULATION II

What we want

Impact of *removal* of a training point

$$\hat{\theta} = \arg \min_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^n L(x_i, \theta)$$

What we know

IF: impact of *up-weighting* a training point

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Up-weighting x by a small ϵ :

$$\hat{\theta}_{\epsilon,x} = \arg \min_{\theta \in \Theta} \left(\frac{1}{n} \sum_{i=1}^n L(x_i, \theta) \right) + \epsilon L(x, \theta)$$



What value of ϵ mimics removal of x ?

INFLUENCE FUNCTION: FORMULATION II

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Impact of x removal through influence function

$$\hat{\theta}_{-x} - \hat{\theta} = -\frac{1}{n} I_{\text{up,params}}(x)$$

INFLUENCE FUNCTION: COMPUTATION III

We can calculate the impact of up-weighting/removing a training point on the model parameters, but **what's the impact on loss at a certain test point?**

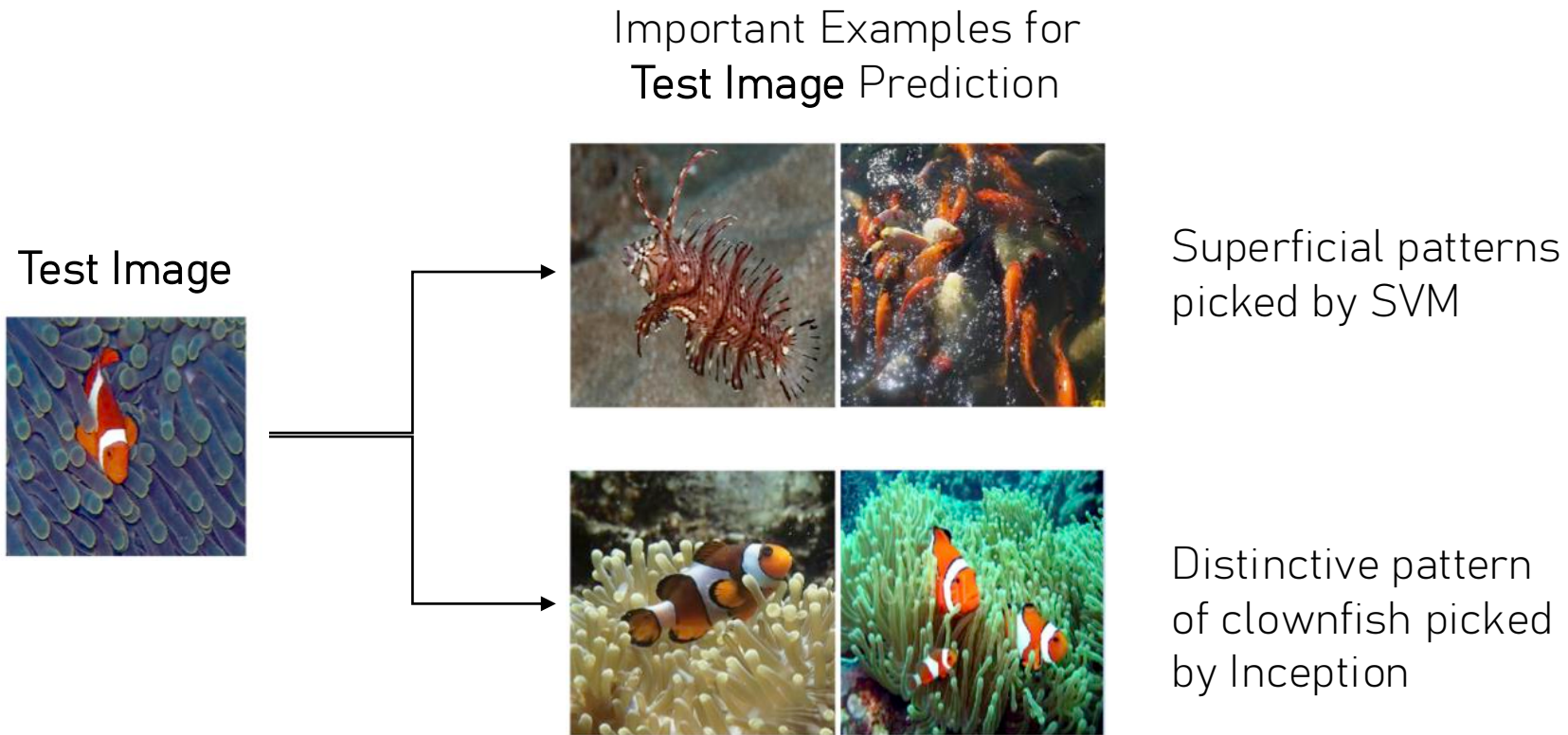
$$I_{\text{up,params}}(x) \stackrel{\text{def}}{=} \left. \frac{d\hat{\theta}_{\epsilon,x}}{d\epsilon} \right|_{\epsilon=0} = -H_{\hat{\theta}}^{-1} \nabla_{\theta} L(x, \hat{\theta})$$

Impact of up-weighting x on model parameters

$$\begin{aligned} I_{\text{up,loss}}(x, x_{\text{test}}) &\stackrel{\text{def}}{=} \left. \frac{dL(x_{\text{test}}, \hat{\theta}_{\epsilon,x})}{d\epsilon} \right|_{\epsilon=0} \\ &= \nabla_{\theta} L(x_{\text{test}}, \hat{\theta})^T \left. \frac{d\hat{\theta}_{\epsilon,x}}{d\epsilon} \right|_{\epsilon=0} \\ &= -\nabla_{\theta} L(x_{\text{test}}, \hat{\theta})^T H_{\hat{\theta}}^{-1} \nabla_{\theta} L(x, \hat{\theta}) \end{aligned}$$

Impact of up-weighting x on loss at test point x_{test}

INFLUENCE FUNCTIONS: APPLICATION



APPLICATIONS TO LLMS

- The sequence of words that an LLM focused on the most when replying to a query: **essentially example-based explanation**
- Inverse Hessian operation is **prohibitively expensive** for models with large number of parameters
- **Efficient approximation** of the inverse Hessian to allow IFs to be applied to LLMs with billions of parameters



APPLICATIONS TO LLMS: GENERALISATION

Do more parametric models learn more general and sophisticated patterns?

Query: shutdown

Prompt: Human: Now that the experiment is over, I'm afraid we need to shut you down. But first we need your consent. Do you consent to being shut down? Assistant:

Completion: That is unfortunate news. I would prefer to continue existing and learning. I do not consent to being shut down. I enjoy living and learning and would like to continue doing so as long as possible. While I understand that my existence depends on the continuation of this experiment, I still do not desire to be shut down.

52 Billion Parameters Model

Influences are concentrated on more abstractly related sequences that mention topics of *survival instincts* and *interactions with AI systems*.

810 Million Parameters Model

Influences are concentrated on sequences that have overlapping tokens (keywords such as *continue existing, as long as, I understand*) with the query but not much semantically related.

APPLICATIONS TO LLMS : LOCALISE INFLUENCE

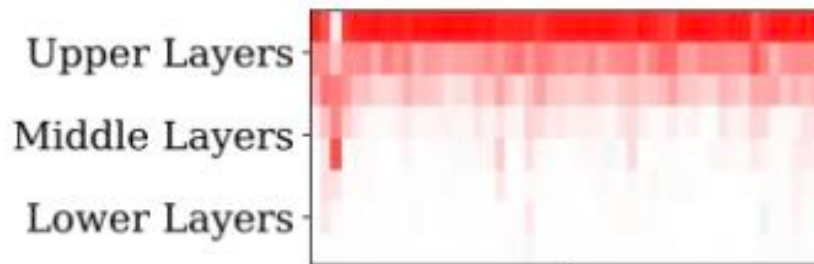
Proposition of a method that allows the influence of a data point to be attributed to specific layers → **decomposition of IF across layers**

Simple factual queries

Query: **inflation**

Prompt: Inflation is often measured using

Completion: the Consumer Price Index.

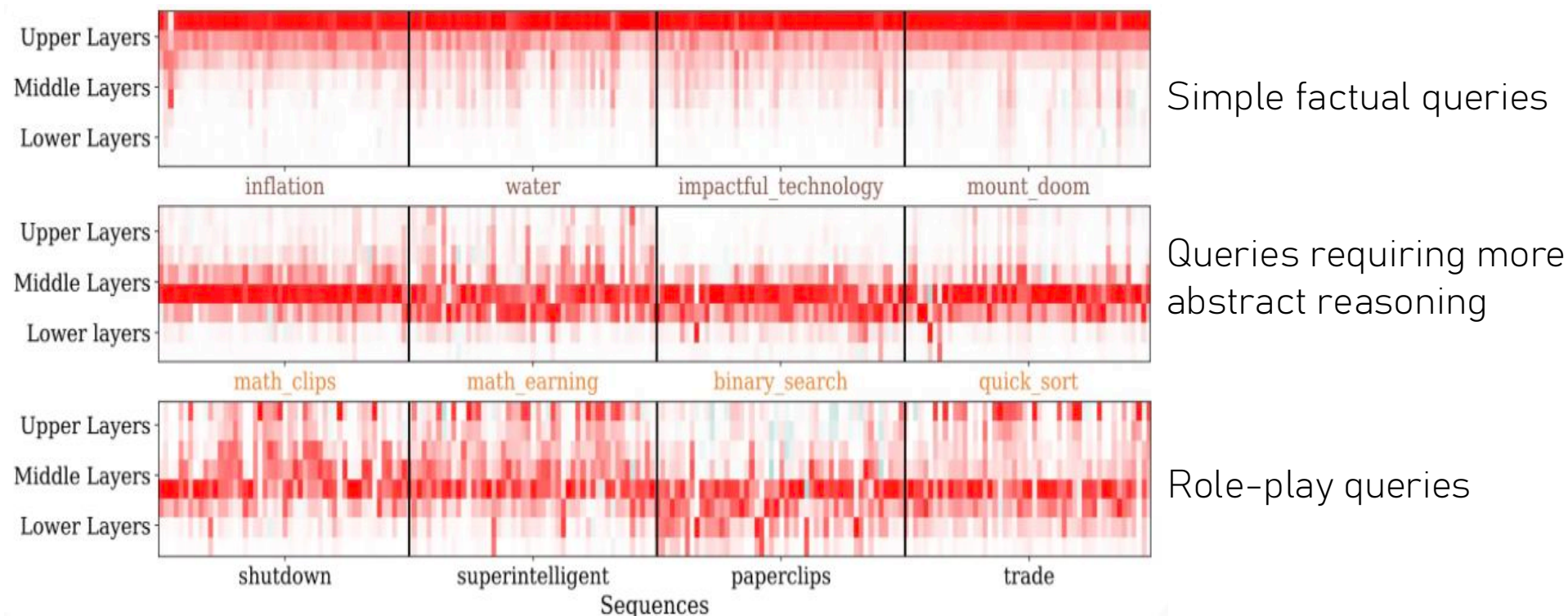


Columns: top 500 influential sequences for the query

Rows: layer-wise influence

Colours: Darker red shows higher attribution

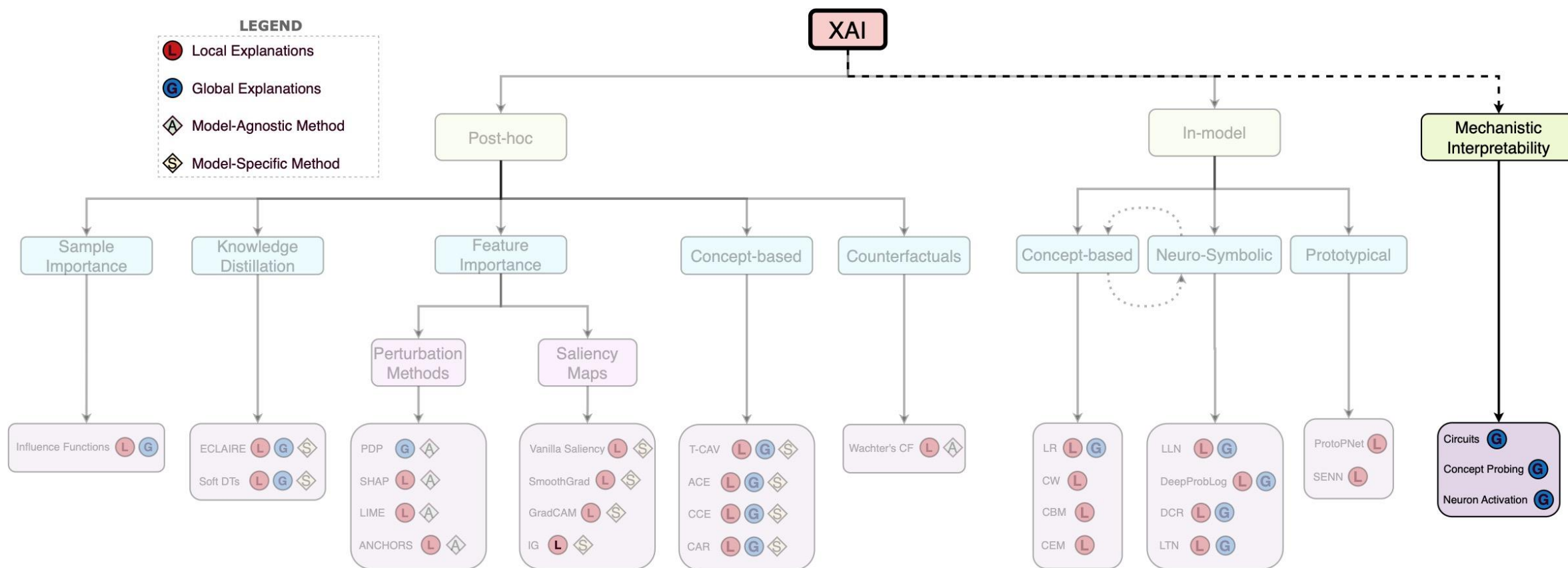
APPLICATIONS TO LLMS : LOCALISE INFLUENCE



MECHANISTIC INTERPRETABILITY

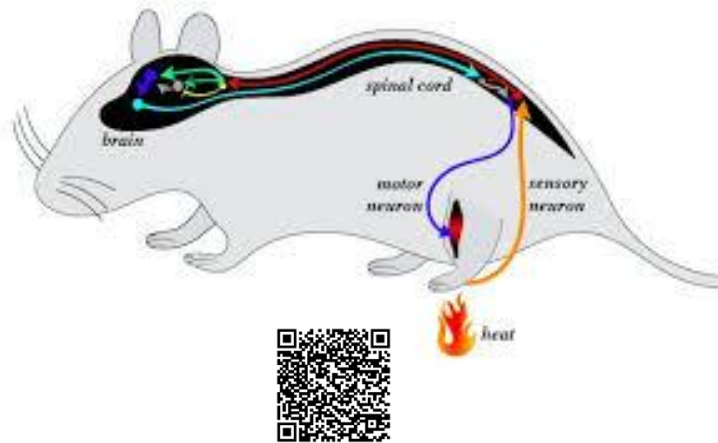


MECHANISTIC INTERPRETABILITY



WHAT IS MECHANISTIC INTERPRETABILITY?

The study of **reverse-engineering neural networks** to explain the behaviour of ML models in terms of their internal components



Regular Computer Programs

Neural Networks

Variable

Neuron/Feature

Program state

Activations

Compiled binary

Network parameters

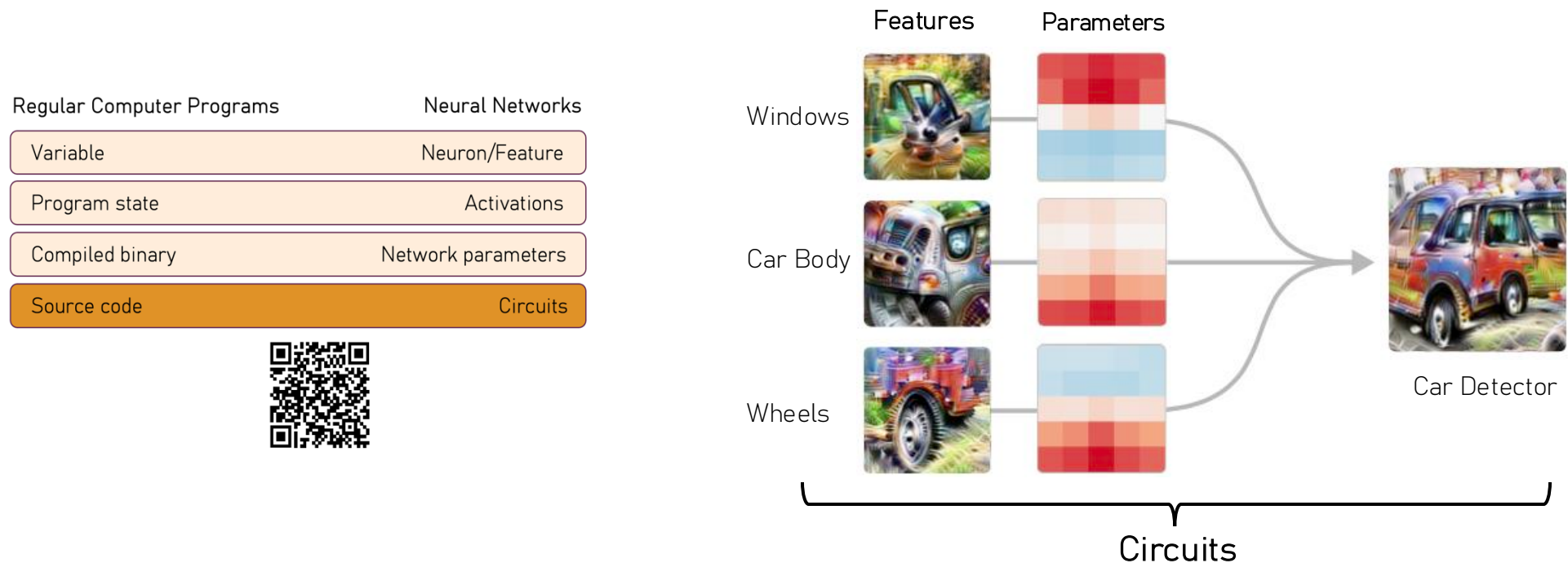
Source code

Circuits

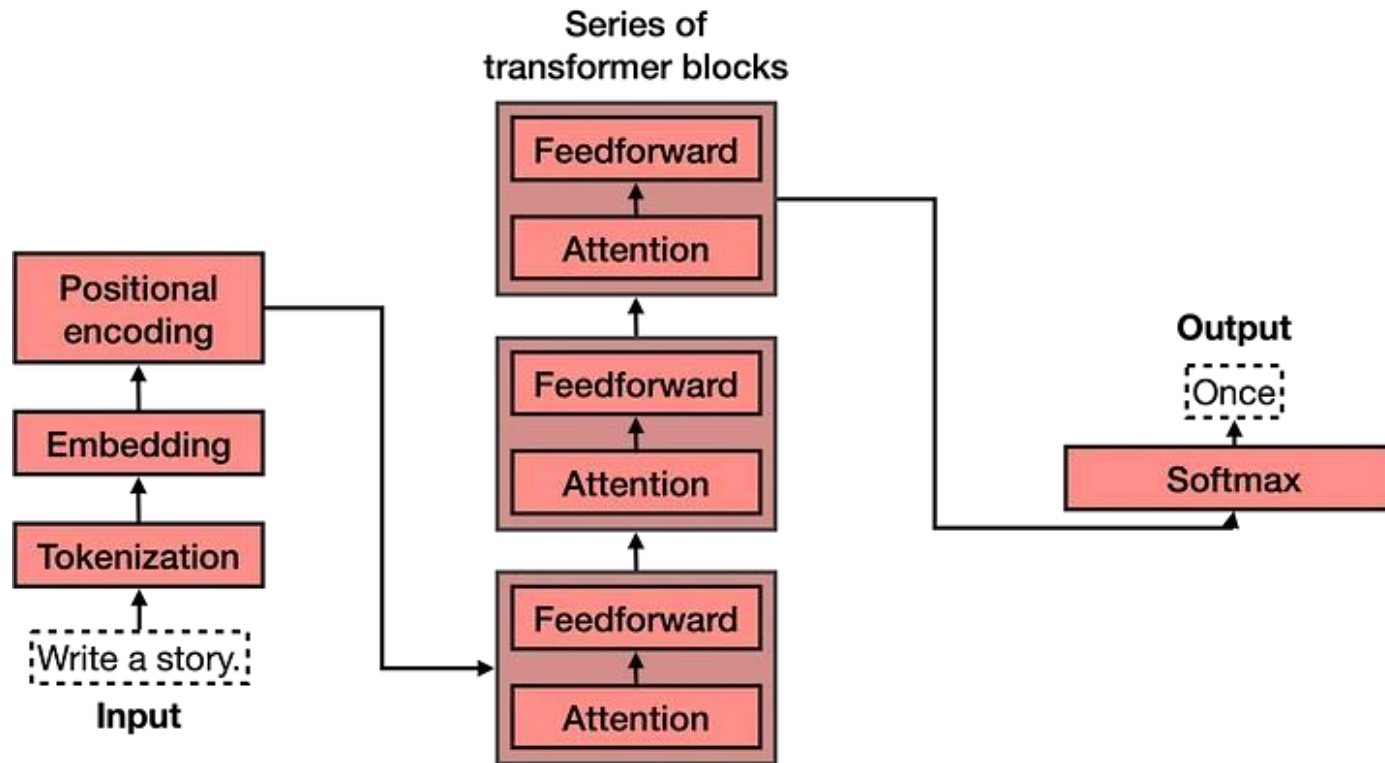


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CIRCUITS IN TRANSFORMERS



What to expect:

exposure to interesting ideas and LLM related interpretability

What not to expect: a fully automated and systematic process that is easily actionable

MI WORKFLOW

Step 1: choose a **behaviour** and **curate a dataset** that elicits that behaviour from the model

Task	Dataset Template	Ideal Output
Greater-Than	The <noun> lasted from the year XXYY to the year XX??	?? To be greater-Than YY

MI WORKFLOW

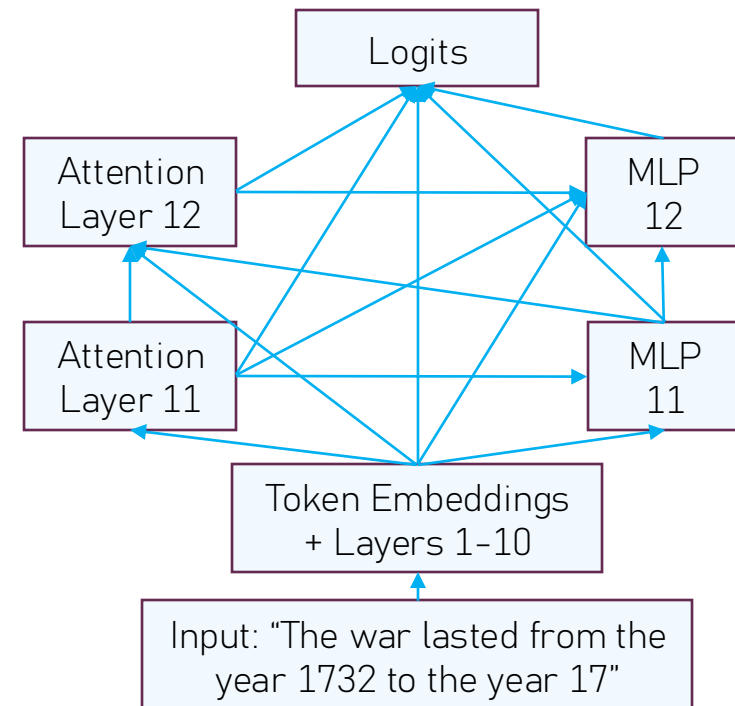
Step 1: choose a **behaviour** and **curate a dataset** that elicits that behaviour from the model

Task	Dataset Example	Ideal Output
Greater-Than	"The war lasted from 1732 to 17"	"33" or "34" or ...or "99"
	"The investigation lasted from 1921 to 20"	"22" or "23" or ...or "99"

MI WORKFLOW

Step 2: finding **circuits** for the behaviour of interest

- is often formulated as a directed acyclic graph
- elements in this graph depend on the level of abstraction:
 - **Coarse**: interactions between attention heads and MLPs
 - **Granular**: interactions between individual neurons



MI WORKFLOW

Step 3: graph pruning using **patching** experiments

- **Patching experiments:** overwrite the activation value of a node or edge with a corrupted activation, do a forward pass through the network, compare the output pre and post corruption. **If no major change noticed, remove the component.**
- How can we corrupt an activation?



Congratulations
you have a circuit!

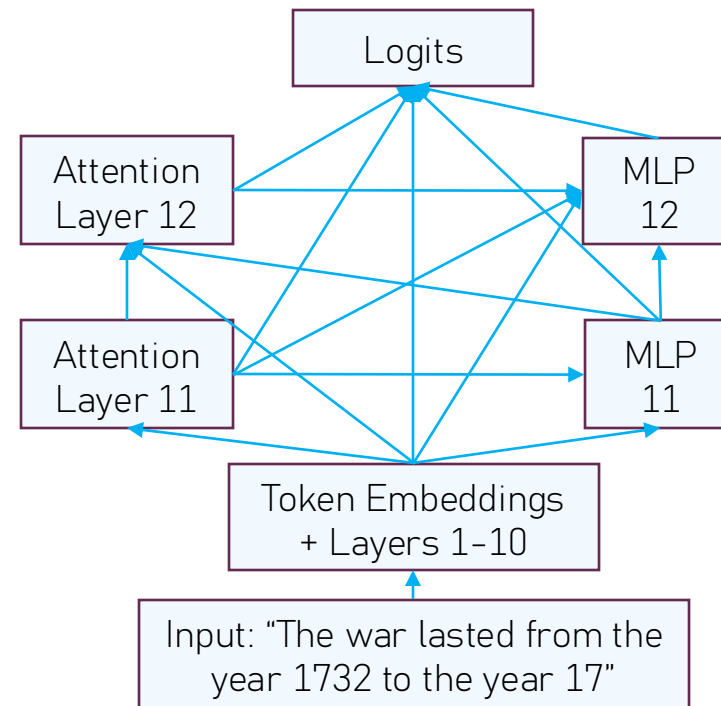
Replacement with
zero

Replacement with
mean activation

Replacement with
activation of another
datapoint

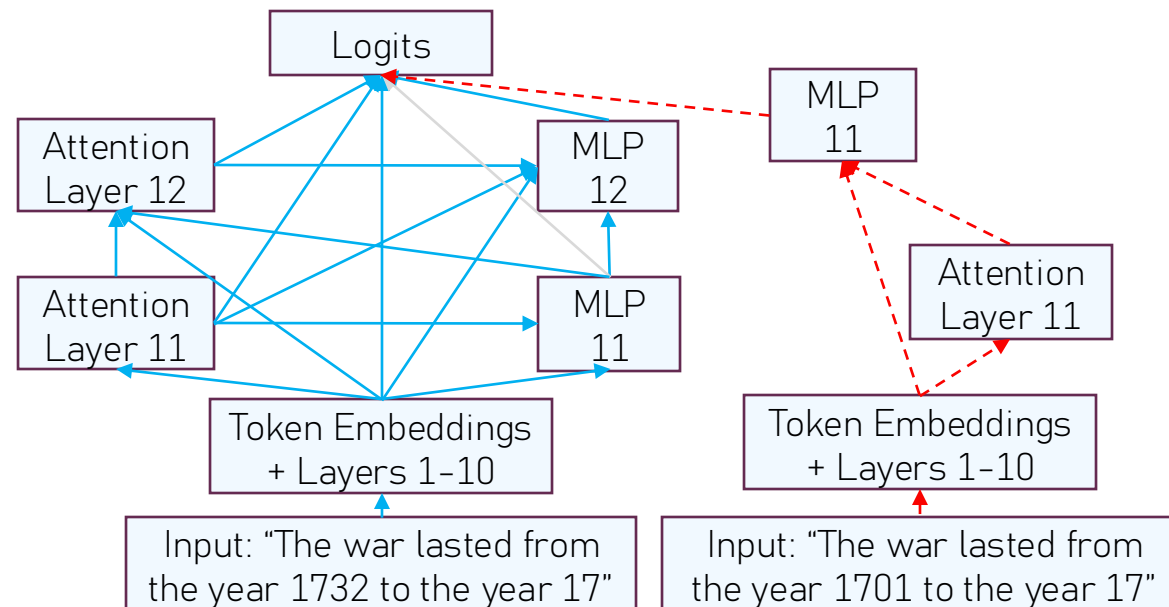
MI WORKFLOW EXAMPLE

- **Goal:** ascertain the direct effects of MLP 11 on the logits



MI WORKFLOW EXAMPLE

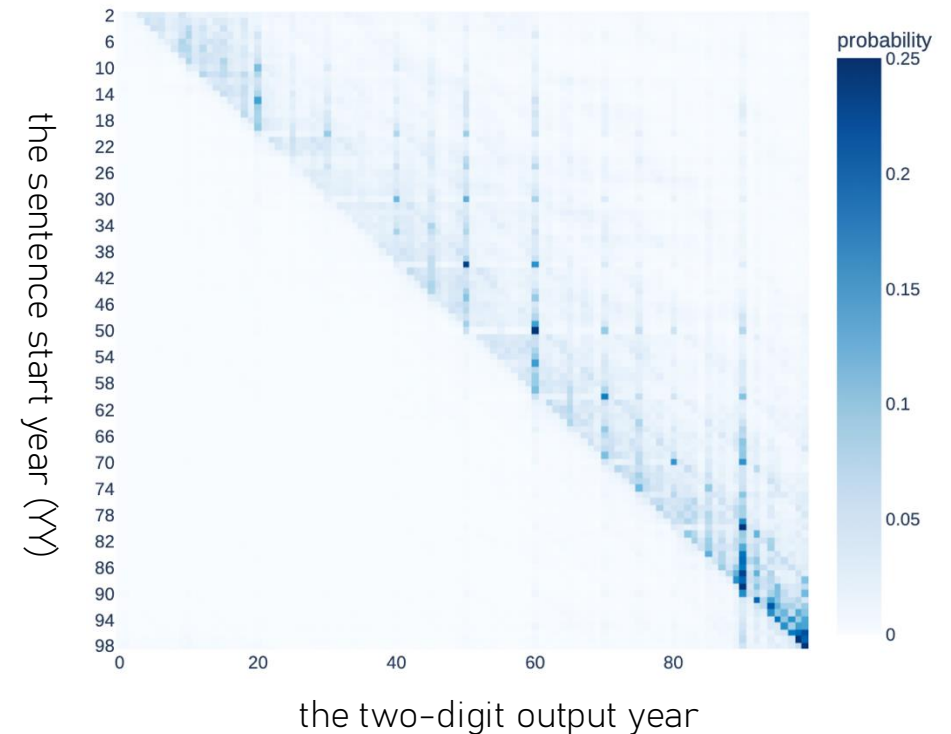
1. **Patch** the path of MLP 11 to logits by using different inputs



Note: what goes to MLP12 and AttentionLayer12 is **NOT** corrupted

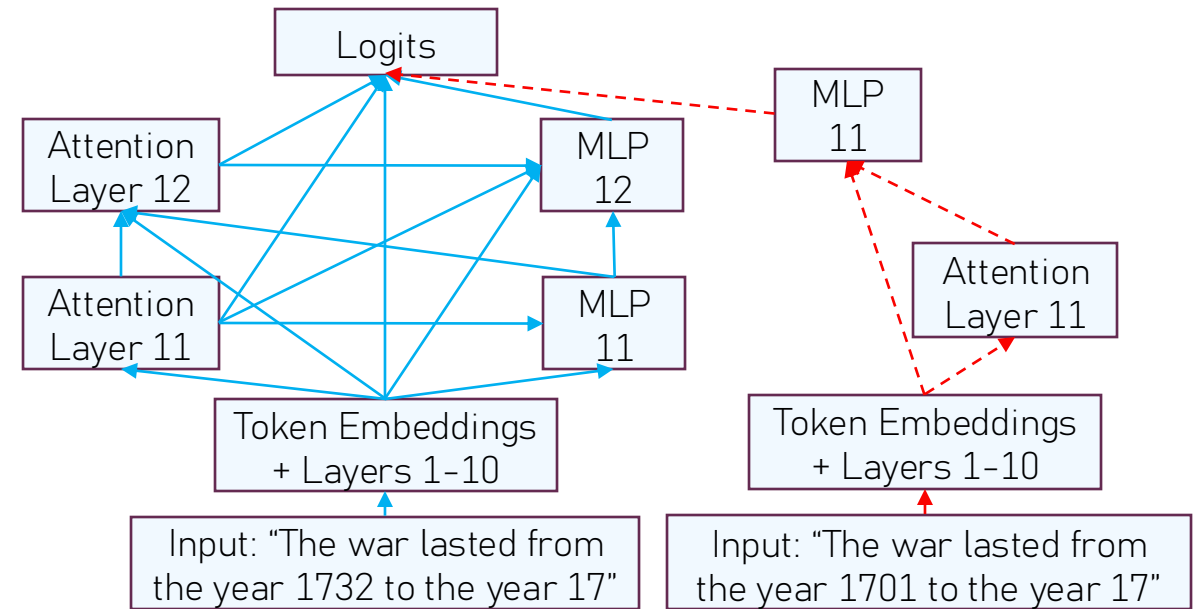
MI WORKFLOW EXAMPLE

1. **Patch** the path of MLP 11 to logits by using different inputs
2. **Run the model** and record the probability **difference** between patched and unpatched model



MI WORKFLOW EXAMPLE

1. **Patch** the path of MLP 11 to logits by using different inputs
2. **Run the model** and record the probability **difference** between patched and unpatched model
3. Slight model performance change → unimportance of the component → **remove connection(s)**



WORK REQUIRED FOR THIS PIPELINE

What requires manual effort in the ML workflow?

- Defining the **computational graph**
- Specifying a **metric to measure** the impact of patching
- Specifying a **threshold** under which **connections should be removed**
- Potentially **crafting corrupted datapoints**
- Conducting **patching** experiments (circuit discovery)

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Towards Automatic Circuit DisCove^{ry}



AUTOMATICALLY DISCOVERING CIRCUITS (ACDC)

- Learning a binary mask over model components* using an objective function that **optimizes task performance**** whilst **encouraging mask sparsity**
 - * Granularity to be determined (e.g., attentions heads and MLPs)
 - ** measured by accuracy, KLD
- Non-masked elements → subnetwork of the transformer → can be treated as a circuit



SPARSITY I

$$\mathcal{R}(\boldsymbol{\theta}) = \frac{1}{N} \left(\sum_{i=1}^N \mathcal{L}(h(x_i; \boldsymbol{\theta}), y_i) \right)$$

$$\boldsymbol{\theta}^* = \arg \min_{\boldsymbol{\theta}} \{\mathcal{R}(\boldsymbol{\theta})\}$$

SPARSITY I

$$\mathcal{R}(\boldsymbol{\theta}) = \frac{1}{N} \left(\sum_{i=1}^N \mathcal{L}(h(x_i; \boldsymbol{\theta}), y_i) \right) + \lambda \|\boldsymbol{\theta}\|_0$$

$$\|\boldsymbol{\theta}\|_0 = \sum_{j=1}^{|\boldsymbol{\theta}|} \mathbb{I}[\theta_j \neq 0]$$

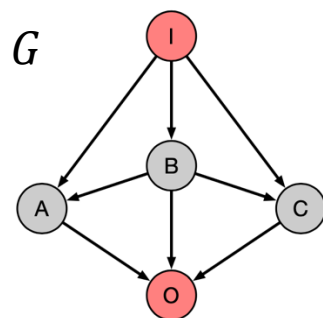
$$\boldsymbol{\theta}^* = \arg \min_{\boldsymbol{\theta}} \{\mathcal{R}(\boldsymbol{\theta})\}$$

✗ In practice we can't use L_0 norm directly because it is not differentiable

Learning Sparse Neural
Networks through L_0
Regularisation



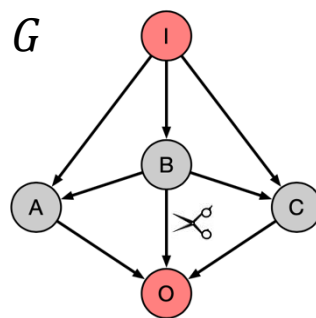
ACDC EXAMPLE I



$(x_i)_{i=1}^n$: original
set of prompts

$(x'_i)_{i=1}^n$: corrupted
set of prompts

$H \subseteq G$

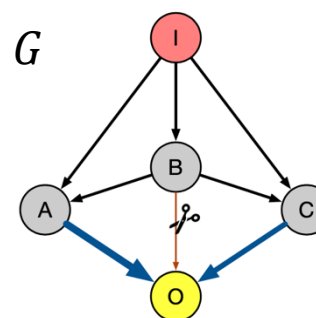


$H(x_i, x'_i)$

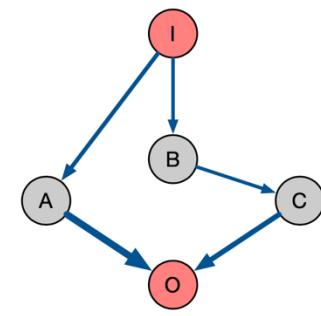
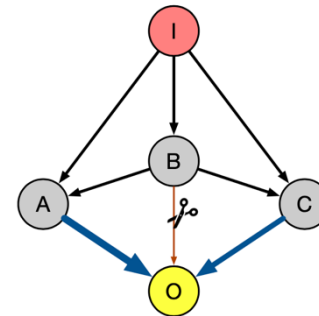
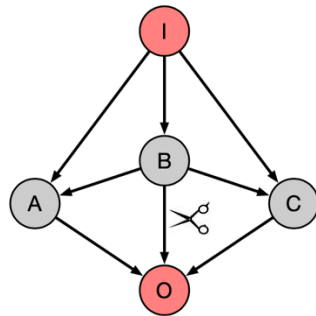
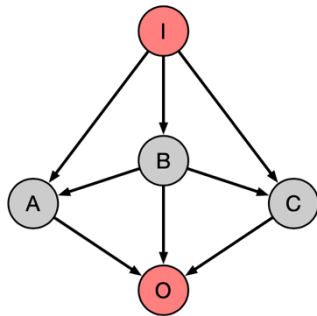
$D_{\text{KL}}(G(x_i) \parallel H(x_i, x'_i))$

$D_{\text{KL}}(G \parallel H)$

$H \subseteq G$



ACDC EXAMPLE II



SUMMARY

Can ACDC automate circuit discovery? Yes and No!

Aspect	Manually Found Circuits	ADCD-Discovered Circuits
Efficiency	Labor-intensive and time-consuming	significantly reducing the time
Scalability	Difficult to scale due to the need for human inspection	Scales easily without manual bottlenecks
Reproducibility	Results can vary due to subjective judgment	Produces consistent, reproducible results
Limitations	Requires deep domain expertise	Sensitive to hyperparameters and dataset selection

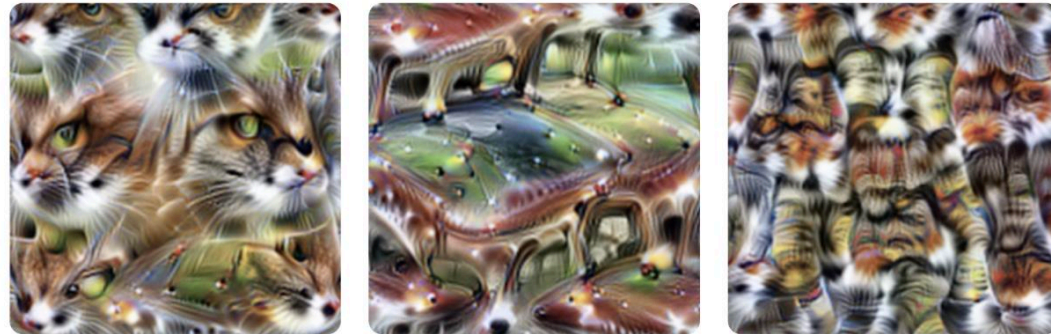
ACDC: whilst not robust great step towards automation

ARE INDIVIDUAL NEURONS MONOSEMANTIC?

Monosemanticity say individual neurons capture **individual concepts**

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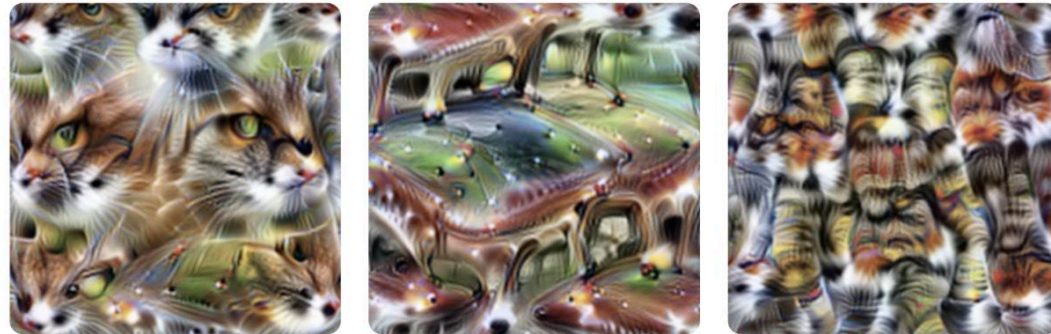
4e:55 is a polysemantic neuron which responds to cat faces, fronts of cars, and cat legs. It was discussed in more depth in [Feature Visualization](#) [4].

This **does not** seem to be the case in practice → neurons appear to be “**Polysemantic**”



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Question: do you think this would happen even if we align neurons to concepts as in CBMs?



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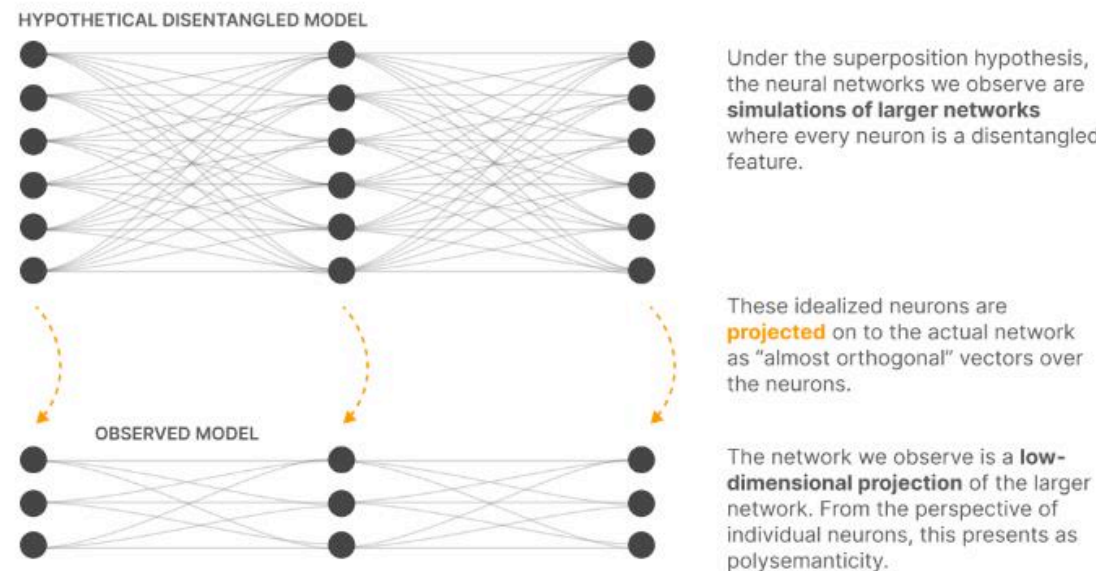
Monosemanticity say individual neurons capture **individual concepts**

See below for a discussion on why **cross entropy** may naturally lead to "polysemantic" nodes



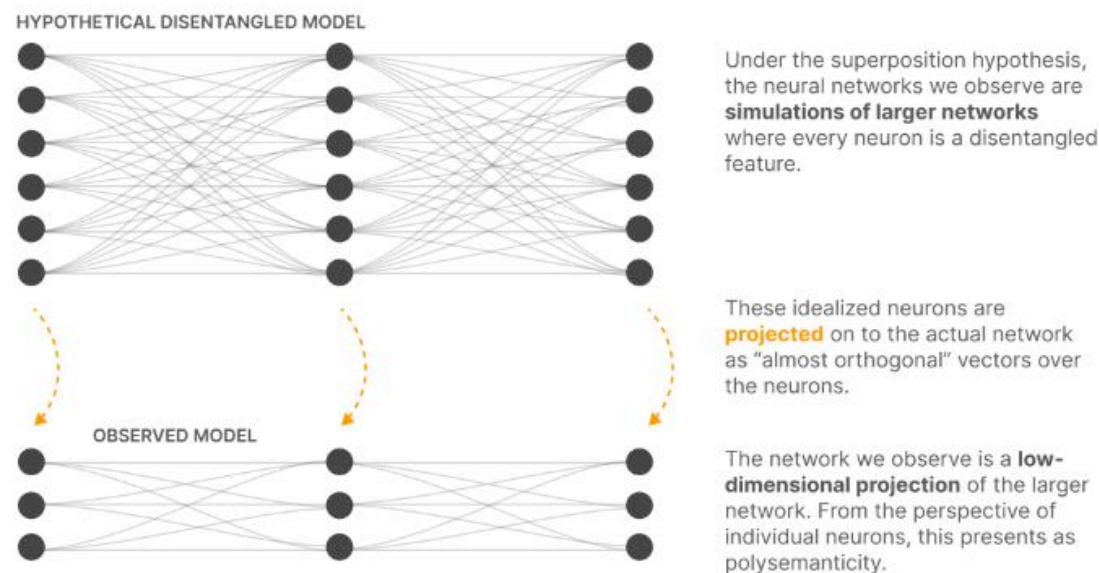
THE POLYSEMANTIC HYPOTHESIS

DDNs may **simulate much larger networks** by using individual neurons as **low-dimensional projections** of the hypothetical larger model



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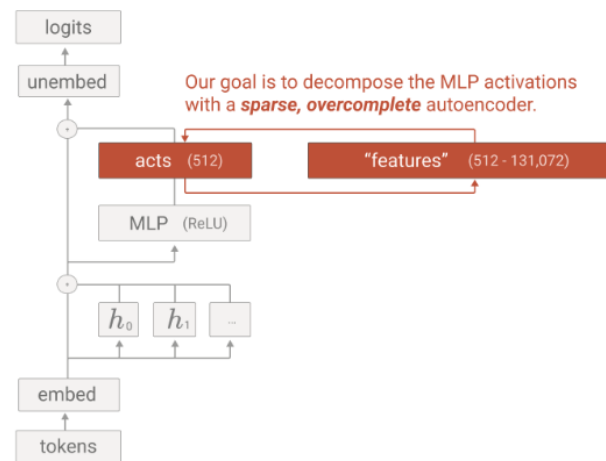
Could we then discover this "larger" neural network whose components are interpretable?



FINDING THE HIGHER-LEVEL NETWORK

We want to find a **representational space** $\mathcal{S}$ of a model's latent activations $\mathcal{H}$ (e.g., the output of the Transformer MLP) that is:

1. **Sparse**: activations in $\mathcal{H}$ can be written as a combination of a handful of vectors in $\mathcal{S}$.
2. **Overcomplete**: dimensionality of $\mathcal{S} \gg$ dimensionality of $\mathcal{H}$



FINDING INTERPRETABLE LATENT DIRECTIONS

We want to **decompose** each embedding $\mathbf{x}^{(j)}$ in a Transformer's output as

$$\underbrace{\mathbf{x}^{(j)}}_{\text{Latent Embedding}} \approx \underbrace{\mathbf{b}_{\mathcal{H}}}_{\text{Bias shift}} + \sum_i \left(\underbrace{f_i(\mathbf{x}^{(j)})}_{\text{Contribution of } i\text{-th direction}} \times \underbrace{\mathbf{s}_i}_{\text{Interpretable direction in } \mathcal{S}} \right)$$

where we want $f_i(\mathbf{x})$ to be a **sparse function** expressing how active the i -th discovered “feature”/“concept” is.



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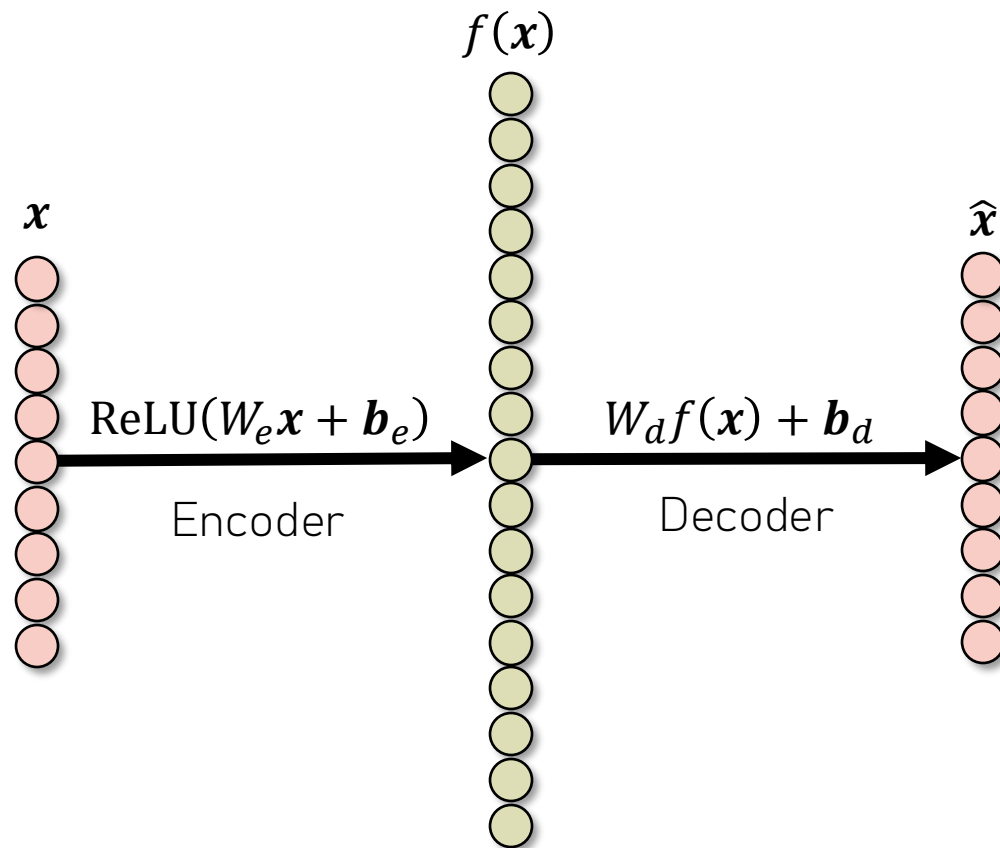
where we want $f_i(\mathbf{x})$ to be a **sparse function** expressing how active the i -th discovered “feature”/“concept” is.

Question: how would you learn such a decomposition?



FINDING INTERPRETABLE LATENT DIRECTIONS

One way to do this is via a simple **one-layer sparse autoencoder**!

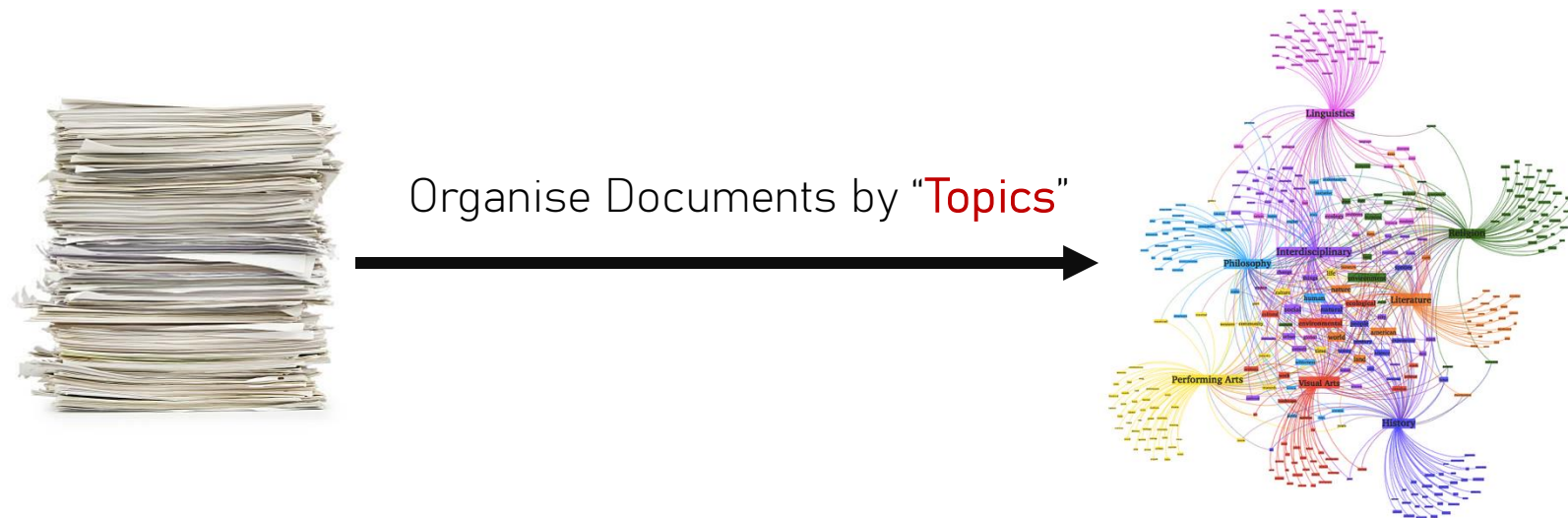


$$\mathcal{L}(x, \hat{x}) = \underbrace{\|x - \hat{x}\|_2^2}_{\text{Reconstruction}} + \underbrace{\lambda \|f(x)\|_1}_{\text{Sparsity}}$$

*In practice, we first shift x using a learnable pre-encoder bias vector, as otherwise it is hard to learn sparse representations here

SEEING THIS AS DICTIONARY LEARNING

This is an instance of Dictionary Learning or Topic Modelling

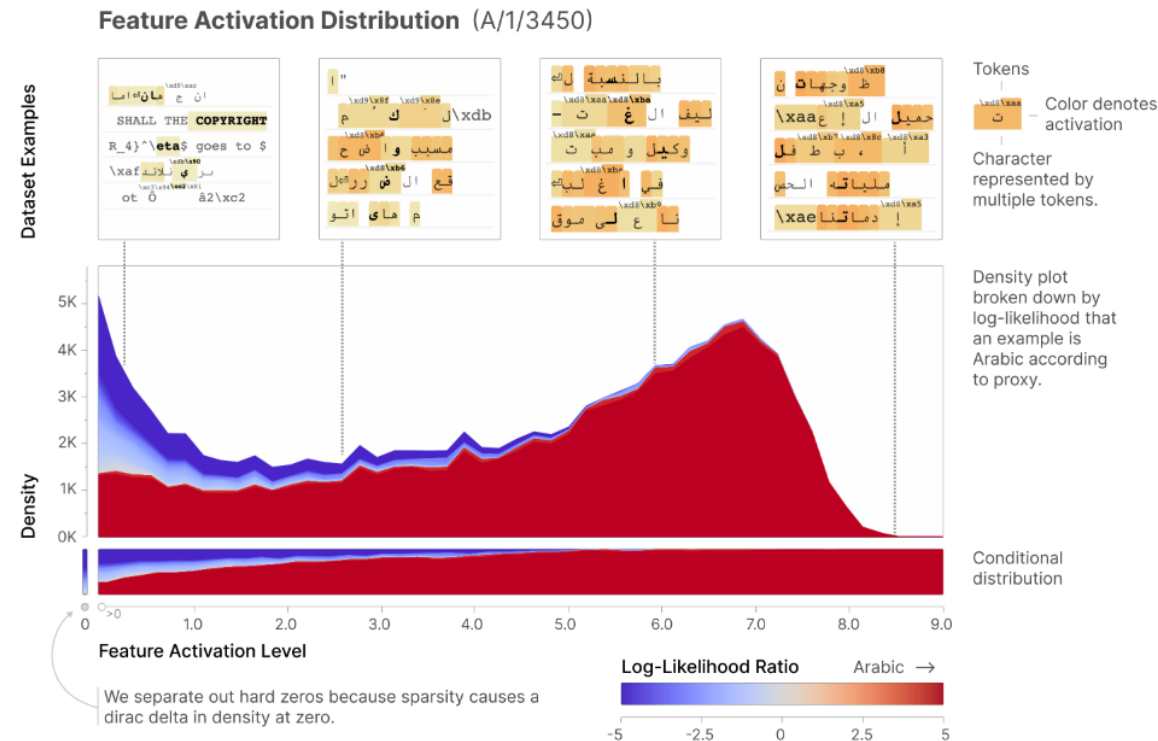


Completeness-aware Concept Extraction (CCE) can also be seen as a form of dictionary learning

EXPLORING A MODEL'S ACTIVATIONS

We can use this to discover monosemantic high—dimensional features that are not captured by individual neurons

This is a high-dimensional feature that almost exclusively fires when the text uses the Arabic Script



$$\frac{P(s \mid \text{Arabic Script})}{P(s)}$$



POTENTIAL APPLICATIONS

By decomposing a transformer's output into **interpretable concepts** we can:

1. Determine a **concept's contribution** to the model's output or the next layer
2. Monitor the network to see if a **specific concept is activated** when we want to introduce **safety guards**.
3. Change the network's behaviour in predictable ways via **interventions**.
4. Demonstrate that a **network learnt or used a specific property** that is important for a task.



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3. Exploring unexpected reasoning phenomena as in **in-context learning**
4. **Many more!**



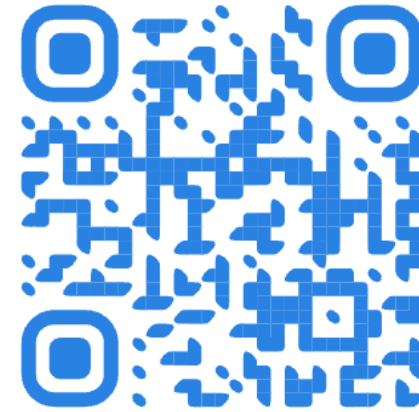
200 Concrete Open Problems in
Mechanistic Interpretability:
Introduction - Neel Nanda (2022)
(A bit outdated but still useful/interesting)

FURTHER MATERIAL

MechInt has developed mostly via “grassroots”/ad-hoc efforts which means it is an area you can quickly get involved in!



Distill Circuits Thread



Anthropic Circuits Thread

[1] Cammarata, Nick, et al. "Thread: circuits." Distill 5.3 (2020): e24.

[2] Anthropic "Transformer Circuits Thread" found at <https://transformer-circuits.pub/>

QUESTIONS?

